SoyMate: viability brief

Is a phone-photo seed grader worth building for France, what would make it work, and where it should go after the soybean proof of concept. Market research done 31/10/2026, during the event; the dataset audits it cites were done before the event and are declared as prior resources. No model results yet: those come from the hackathon build. Detailed notes: `research/viability/`; adversarial review: `research/critique.md`.

How to read the numbers

Each figure carries a source number **[n]** (list at the end) and a tag: **[V]** read on the primary source; **[F]** secondary source or snippet; **[NV]** our own calculation or inference (inputs given); **[A]** an assumption, listed in section 8 and varied in the scenario table.

In one page

What we build at the hackathon. A proof of concept on soybean, the only legume with public labelled seed-defect data: segment every seed in a photo, classify it, give the % of each defect with a confidence interval, and a pass/fail against a selectable food-grade threshold.

Soybean alone is a weak business. French food soy is ~45 kt/yr out of ~400 kt [2] [V], and the food vs feed price gap is "très faible": under 25 €/t in 2019/20 and 2022/23 [10] [V] [11] [F] (~70 €/t quoted for 2021 [6] [V]).

Lentil is where a tonne is worth defending. Food lentil sells for 806 to 873 €/t [24] [V] against ~250 €/t for feed protein crops [23] [V]: ~550 to 620 €/t at stake [NV]. In 2025 the mean bruchid damage per lot was 24.4 %, the highest of the survey's 2021-2025 series (40 lots measured) [15] [V]. An estimated 45 to 55 % of the lentils eaten in France are imported [13] [V] [NV, production is estimated].

The market exists. Cropify already sells AI lentil and chickpea grading in Australia; Grain Trade Australia's Trading Standards Committee judged it worth "commercial consideration" in August 2026 [26] [V]. Nobody publicly covers French soy or pulses: Inarix, present in ~600 silos, covers cereals only [28] [29] [V].

Where the value is, and where it is not. A better grade mostly moves money between farmer and collector; it creates value only when it **changes a decision** (sort or not, which contract or storage cell, accept or contest a lot) or **removes a dispute**. And one photo is noisy: at 25 % damage a 200-grain count is ±6 points [NV, binomial]. So the product is **a fast, repeatable, documented grade that the farmer can check**, built from enough grains, not a "€ per point" saving.

Money (section 6, illustrative). At 1,500 € per site per year [A4], a collector breaks even with any one of: ~11 avoided lab arbitrations, 375 to 750 replaced manual counts, or one 28 t lentil lot correctly kept in the food outlet every ~10 years. Customer return: 0.3x without allocation gains, ~1.9x in our base case, 7.2x optimistic [NV, A10-A12]. Our marginal cost per analysis is near zero (phone, CPU model); the fixed cost is French data and a pilot.

Impact. In 2025 the lentil sorting loss was up to ~5 to 8 M€ [NV, upper bound], carried by farmers through deductions they cannot verify today. A grade they can check before and at delivery supports their income, the Track 3 question [1]. It does not raise demand or acreage, and we do not claim it does.

Honest limit. France alone is a small market (a few M€ of value per year across soy and pulses [NV]). The viable route is a phone-only soy and pulse module for a tool already at the intake desk, or EU expansion, with lentil first.

1. What the hackathon proof of concept is, and why soybean

- **Only soybean has the data.** We found no public RGB photo dataset of damaged pea, lentil, faba bean, chickpea or beans after two searches (Mendeley, Zenodo, figshare, Kaggle, Hugging Face, GitHub, Data in Brief, Scientific Data) [43] [NV, negative search]. The closest French data is Radio_lentille (Terres Inovia): 807 X-rays of bruchid-damaged lentils with count spreadsheets (MIT licence) [43]. X-rays, not phone photos, but its counts give a real distribution of lot damage.
- **Data we use:** Indian soybean beetle-damage set (audited before the event, declared as a prior resource: 80 to 84 % of the published validation and test images leak from train, so the split is regenerated at the event), Chinese 5-class Soybean Seeds (intact, spotted, immature, broken, skin-damaged), and three sets of real multi-seed photos: Ghana phone photos (~150 seeds each), mechanical-harvest images with intact/crushed/impurity masks, Indonesian flatbed scans. All CC BY 4.0 [43].
- **What the PoC will measure** (results come during the event): R1 leaky vs clean split; R2 per-class F1 on the clean test set; R3 transfer from one dataset to another with 0, 5, 10, 25, 50 labelled grains per class (the cost of a new capture setup); R4 seed-count error on real multi-seed photos and lot-level error on simulated lots (declared synthetic).

- **What R3 does and does not show:** it measures how much labelled data a new capture setup costs within one species. It says nothing direct about lentil, whose shape and seed coat differ (a "stained" class trained on soy may fire on naturally speckled lentils). Everything about lentil below is a hypothesis until it is tested on French lentil photos.
- **Soy classes that matter in France:** green/immature seeds (the 2025 food-limiting defect [3]), broken, stained. The "insect hole" class comes from an Indian storage pest that is not a French soy problem [CONTEXT.md §3]; it is kept only as a method demo for the lentil route, where bruchid exit holes are the main defect.

2. The French market in numbers

	Soybean	Lentil	Pea	Faba bean
Area	150 kha (2025), down from 187 kha in 2020 [5] [V]	34,216 ha (2024) [13] [V]	123.5 kha (2024) [13] [V]	83.4 kha (2024) [13] [V]
Production	~392 kt (2025) [5] [V]; 298 kt forecast for 2026 [9] [F]	40 to 59 kt (2024), no official figure [NV, A5]	350 kt (2024) [13] [V]	224 kt (2024) [13] [V]
Human food	45 kt/yr, ~11 %, almost all French, under contract [2] [8] [V]	Almost all; an estimated 45 to 55 % of consumption imported [13] [V] [NV, A5]	136 kt food and non-food ingredients (2024/25), mostly fractionation [13] [V]	11 kt (2024/25) [13] [V]
Producer price	596 to 617 €/t conventional (2022/23) [11] [F]	806 to 873 €/t [24] [V]	250 feed vs 280 food €/t, indicative [25] [V]	200 to 260 €/t (feed range) [23] [V]
Quality problem	2025: 37 % of lots >1 % green (immature) seeds vs 11 % in 2024, "pénalisant" for food; protein 39.8 % DM, a 10-year low [3] [V]	2025: mean 24.4 % bruchid-damaged grains per lot (40 lots), highest of 2021-2025; 36.8 % in Centre-Ouest [15] [V]	2025: only 55 % of lots under 1 % broken grains, the most limiting food criterion [44] [V]	Egypt food exports fell from 245 kt to 87 kt, mainly over bruchids [21] [F]
Visible in a photo?	Green seeds yes; protein no (needs NIR)	Exit holes yes; larvae inside no	Broken yes	Holes yes; larvae no

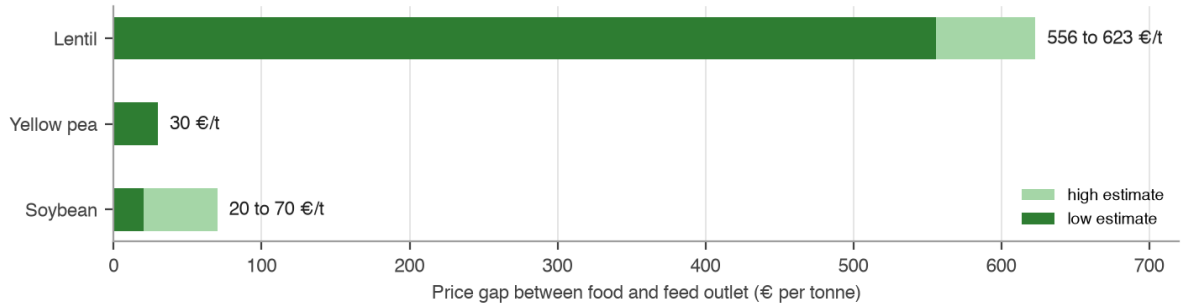


Figure 1. Food minus feed price per tonne. Soybean: observatory gap 2019-2023 (low) [10] [11] and the 2021 contract premium (high) [6]. Pea: indicative prices 280 vs 250 €/t [25]. Lentil: producer price 806 to 873 €/t [24] minus a 250 €/t feed proxy [23] [A2], because no feed-lentil price is published [NV].

3. Why soybean alone is not a business

- **Thin premium.** The interprofession's producer-price observatory calls the soy food vs feed gap "très faible" [10] [V]: 343 vs 326 €/t (2019/20), 617 vs 596 €/t (2022/23) [11] [F]. The whole French food-soy premium is ~0.9 to 3.2 M€/yr [NV: 45 kt x 20 to 70 €/t]. A 28 t lot wrongly sent to feed would lose ~560 to 1,960 € [NV, A1, A6], but food soy is grown under contract with set varieties and we found no record of how often that happens.
- **Few buyers, who already control quality.** Six soyfood plants [6] [V]; Olga alone buys 8 to 9 kt/yr [12] [V]. They use variety lists, contracts and their own sorting [6] [12] [V].
- **No public French food spec** for broken, split or stained soy; contracts set operator-specific limits on green, black, broken and half seeds and hilum colour [7] [V]. The national survey does not measure broken or stained soy [3] [V].
- **Shrinking supply:** area -20 % since 2020 [5] [V]; 2026 forecast 298 kt (-25 %) [9] [F].
- **What still works for soy:** green (immature) seeds were the visible defect that hurt the food outlet in 2025 [3] [V], and "immature" is one of our classes. That makes soy a credible first use, not a business on its own.

4. Why lentil is the extension worth building

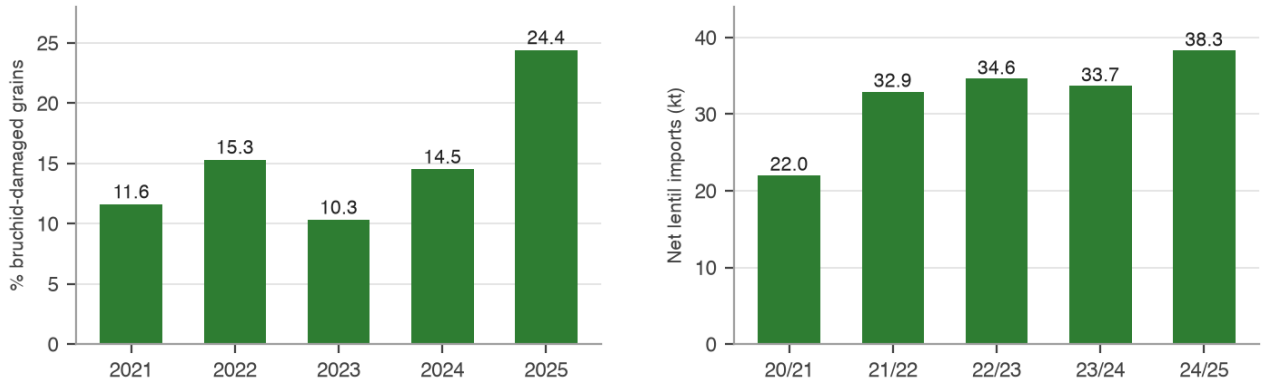


Figure 2. Left: mean % bruchid-damaged grains per lentil lot, national quality survey (2025: 40 lots measured) [15] [V, 2021-2024 read from the survey chart]. Right: France's net lentil imports, campaigns July-June [13] [V].

- **High value per tonne:** ~550 to 620 €/t between food and feed [NV, 24, 23, A2], about 10 to 30 times the soy gap.
- **A growing problem with no field fix:** 2025 had the highest mean damage of the survey's 2021-2025 series [15] [V]; "la lutte chimique préventive est actuellement dans une impasse" [20] [V].
- **Far outside export grades:** under Canadian Grain Commission grades, total damage above 10 % already falls below grade No. 3 [41] [V], so a 24 % lot is far from any food grade before sorting [NV].
- **Import substitution:** net imports grew from 22,030 t (2020/21) to 38,282 t (2024/25), about half of them from Canada [13] [V]. Policy names local légumes secs as a priority [19] [V].
- **The decision is sorting and pricing, not food vs feed.** Damaged lentil is cleaned or optically sorted back to standard [20] [V], and the sorter's reject fraction tells the collector the loss by weight, after the fact [NV]. The tool's role is earlier: before unloading and before sorting, to decide whether to sort, which cell or contract the lot goes to, and to give the farmer the same number. At 24 % damage the sorting loss is ~136 € per harvested tonne, ~5 to 8 M€ nationally in 2025 (upper bound) [NV].
- **Proof that buyers pay:** Cropify grades lentils for GrainCorp, Louis Dreyfus, ETG Wimpak and Shannon Bros, plans up to 25 units for the 2026 harvest, adds chickpea, then faba in 2027; Grain Trade Australia's Trading Standards Committee judged it worth "commercial consideration" [26] [27] [V].

5. Value model: where value is created, and where it is not

A grade is a price. Correcting a grade moves money between farmer and collector, and if manual grading is noisy but unbiased the corrections cancel out across trucks [NV]. So "€ per point of correction" is not a value model. Value is created only in these places:

Source of value	Mechanism	Evidence	Status
Decisions that change	Sort or not; which cell or contract; accept or contest a lot before unloading	Lentil diverted from food loses ~550 to 620 €/t [NV, 24, 23]	How often a decision changes is unknown [A3]
Fewer disputes, same grade everywhere	One method across staff and sites, with stored images as evidence	Australian receival grading varies "from person to person" and between sites [45] [V]; lab arbitration is 130 € HT per lot [37] [V]	Dispute frequency unknown
Speed at the harvest peak	Shorter queues when trucks arrive together	Cropify: ~5.3 min per sample [27] [V]; manual pulse grading "up to 20 minutes" in Australia [46] [F]	Partly sourced
A grade the farmer can check (impact)	Same photo method before delivery and at reception	Track 3 asks how farmers earn from plant protein [1]	Design hypothesis

How many grains a grade needs [NV, binomial 95 % half-width, in percentage points]

True damage	150 grains	200 grains	500 grains	1000 grains	2000 grains
2 %	±2.2	±1.9	±1.2	±0.9	±0.6
5 %	±3.5	±3.0	±1.9	±1.4	±1.0
15 %	±5.7	±4.9	±3.1	±2.2	±1.6
25 %	±6.9	±6.0	±3.8	±2.7	±1.9

- At realistic lentil damage (15 to 25 %), even 1,000 grains leave ±2.2 to ±2.7 points. No method, human or photo, can resolve a 1-point difference on one sample. Phone grading earns its place by counting many grains quickly and the same way every time, not by precision on one photo.
- Certifying a strict threshold is harder still: with zero defects found, the Wilson upper bound is ~2.5 % for 150 grains; showing ≤1 % needs ~380 clean grains [NV]. Classifier errors add bias: a 2 % false-positive rate makes a clean lot read ~3 %, so reported % must be corrected with the measured sensitivity and specificity, and the app must say "inconclusive, add grains" near the threshold.

Scale check, France only [NV]

Quantity	Value	Inputs
Lentil trucks per year	~1,430 to 2,110	40 to 59 kt [A5] / 28 t [A1]
Lentil sorting loss at stake, 2025	~5 to 8 M€ (upper bound)	24.4 % damage [15] x production x ~556 €/t
Food-soy premium pool	~0.9 to 3.2 M€/yr	45 kt [2] x 20 to 70 €/t [10] [6]
Protein-crop collectors	477 (2024/25)	[14] [V]; they handle pea and faba; the number handling lentil was not found
Illustrative revenue if all 477 paid 1,500 €/yr	~0.72 M€/yr	[A4]

Reading: France alone cannot support a standalone company, and the value per site rests on decisions and disputes we have not measured yet [A3]. Viability needs a channel (a module in an existing intake tool) or more markets (EU pulses, more crops), and a pilot that counts how often a grade changes a decision.

6. Money: customer ROI, our revenue, price benchmarks

All figures below are illustrative calculations [NV] on the assumptions in section 8; each input is named so a reader can change it.

6.1 Break-even for a collector paying 1,500 €/yr [A4]

Instead of claiming an ROI we cannot measure yet, we ask: **what would the tool have to do, per season and per site, to pay for itself?** Any one of these is enough:

The tool avoids...	Value of one	Needed per season to break even	Inputs
a manual defect count (labour)	2 to 4 €	375 to 750	10-20 min at ~12 €/h gross [38] [F] [NV]
an arbitration-lab analysis	140 €	11	130 € HT analysis + 10 € handling [37] [V]
a food-soy lot wrongly sent to feed (28 t)	560 to 1,960 €	1 to 3	28 t [A1] x 20-70 €/t [A6]
a sortable lentil lot wrongly kept out of food (28 t)	15,568 to 17,444 €	0.09 to 0.10 (one every ~10 to 12 years)	28 t [A1] x 556-623 €/t [24] [A2]

Reading: labour alone needs hundreds of counts per site, which only large intake sites reach. The case is made by **allocation**: one lentil lot routed wrongly in ten years pays the subscription. Whether that happens is exactly what a pilot must count [A3] [A12].

6.2 Customer ROI in three scenarios (one lentil-receiving site, one season)

	Conservative	Base	Optimistic
Manual counts replaced x 3 € [A10]	150 = 450 €	300 = 900 €	600 = 1,800 €
Arbitrations avoided x 140 € [A11]	0 = 0 €	2 = 280 €	5 = 700 €
Wrong lentil allocations avoided x 16,500 € [A12]	0 = 0 €	0.1 = 1,650 €	0.5 = 8,250 €
Value per season	450 €	2,830 €	10,750 €
Cost (subscription [A4])	1,500 €	1,500 €	1,500 €
Return per € spent	0.3x	1.9x	7.2x

Reading: without allocation gains the tool does not pay for itself; with one avoided lentil misallocation every ten years it roughly pays twice over. The base column is our working hypothesis, not a result. 16,500 € is the midpoint of the 28 t lentil lot value above.

6.3 Our side: price, cost, revenue

- **Price:** 1,500 € per site per year [A4], or credits per analysis as Inarix sells [28] [V]. Per sample we would sit between a manual count (~2-4 € labour [NV]) and imaging competitors (ZoomAgri ~$15 [47] [F]; Nebulaa Rs 150-200 [48] [F]).
- **Marginal cost:** near zero per analysis: the model is frozen features plus a linear layer and runs on a CPU in seconds [NV, our design]; no hardware to ship, unlike Cropify's units [26] and Ground Truth Ag's ~CAD 15k benchtop [31].

- **Real costs are fixed:** two harvests of labelled French photos, calibration against a visual reference count, a pilot with 1-2 collectors, sales. Public calls fund this kind of work: the 2025 legume-chain call gave 11.7 M€ to 10 projects, ~1.2 M€ each on average [42] [V] [NV].
- **France revenue:** 10 % of the 477 protein-crop collectors at 1,500 € = ~72 k€/yr; all of them = ~0.72 M€/yr [NV, A4]. Plus the six soyfood plants as an early, concentrated market [6].
- **So the business is not France alone:** it is (a) licensing a soy and pulse module to an intake-desk incumbent that already sells per site and per credit (Inarix [28] [29]), or (b) the same product in other EU pulse markets, following Cropify's Australia-to-North-America path [26].

6.4 Price benchmarks: what a grade costs today

Option	Hardware	Price	Soy / pulses	Source
Manual count by the intake technician	None	~2-4 € labour, 10-20 min	Yes	[38] [F] [NV]
Arbitration lab (Incograin)	Lab	130 € HT + 10 € handling	Yes	[37] [V]
Ground Truth Ag	Benchtop unit	~CAD 15k + 1.5k/yr	In development	[31] [V]
ZoomAgri	Scanner	~$15 per sample	Soy claimed	[47] [F]
Nebulaa MATT	Device	Rs 150-200 per sample	Pulses (India)	[48] [F]
Cropify	Unit at sample stand	Not public	Lentil, chickpea	[26] [V]
Inarix	Phone	Licence + credits, on quote	Cereals only	[28] [V]
SoyMate	**Phone**	**~1,500 €/yr per site [A4]**	**Soy, then lentil**	Our proposal

7. Competition and market proof

Player	What it does	Soy or pulses?	Price / model
Cropify (AU)	AI grading at receival: defective and discoloured grain, foreign material; digital record per load [26] [V]	Lentil (2025), chickpea (2026), faba 2027; Canada and US next [26] [27] [V]	Not public
Inarix (FR)	Phone photo: variety, protein, broken grains, TKW; ~600 silos, ~40 % of French barley [29] [V]	Cereals only publicly [28] [V]; soybean variety R&D; in Canada [29] [V]	Licence per user + credits [28] [V]
Ground Truth Ag (CA)	Benchtop vision + NIR, 1-2 kg sample [30] [V]	Peas, lentils, soy, faba "in development" [30] [V]	~CAD 15k + 1.5k/yr [31] [V]
Cgrain, FOSS EyeFoss	Lab or intake imaging of grains [32] [33] [V]	Small grains only	Quote
SeedCount	Flatbed seed imaging [34] [V]	Old lentil module	Quote
ZoomAgri (AR)	Scanner imaging: barley purity, physical quality of grains [47] [V]	Soy claimed in press, not verified [47] [F]	~$15 per sample (press) [47] [F]
Nebulaa MATT (IN)	Image analysis of grains and pulses, ~1 min [48] [F]	Pulses (India)	Rs 150-200 per sample [48] [F]
Optical sorters (Bühler, TOMRA)	Sort pulses at packers	Yes, but they sort, not grade	Capital sale

Threats: Cropify is the same product, already recognised, expanding abroad. In France, Inarix already has the silos, phone capture, the licence-plus-credits model and the same DINOv2 backbone [29]; adding pulses would be a small step. **No technical moat:** "not in France yet" is a timing gap. **Differentiator:** phone only, no hardware, where Cropify installs units at sample stands [26] and Ground Truth Ag sells a benchtop unit [31]. **Realistic channel:** a pulse and soy defect module licensed to, or built with, an intake-desk incumbent.

8. Assumptions register

Id	Assumption	Value used	Basis	How to check
A1	Truck size	28 t	Common French semi-trailer load [A]	Ask a collector
A2	Value of a lentil tonne outside the food outlet	~250 €/t	Feed protein-crop range 200 to 260 €/t [23] [V]; no feed-lentil price published	Collector or packer price for downgraded lentil
A3	A faster, repeatable grade changes some decisions (sort, cell, contract) or prevents some disputes	Not quantified	Manual grading varies between people and sites [45] [V]; no French figure on how often a grade changes a decision or is contested	Pilot: count decisions changed and disputes per 100 lots
A4	Subscription price per site	1,500 €/yr (illustrative)	Order of magnitude of Ground Truth Ag's CAD 1,500/yr fee [31] [V]; Cropify price not public; no payback is claimed	Interviews with 3 collectors
A5	Lentil production	40 to 59 kt	34,216 ha [13] x 1.16 to 1.71 t/ha [18] [15] [NV]	Official tonnage (none found for 2024-25)
A6	Soy food premium	20 to 70 €/t	[10] [11] [6]	Current contracts
A7	% of damaged grains = % of lot weight lost to sorting	1:1	Simplification: holed grains weigh less, so the 5 to 8 M€ sorting loss is an upper bound [NV]	Weigh sorted fractions
A8	Contracts price visible damage	Yes	Visual counts are the reference method [36] [V]; infestation needs dissection	Read Incograin addendum 7 bis (members only) [40]
A9	Transfer cost measured on soy (R3) predicts French lentil	Hypothesis	No public lentil defect data to test it	Label ~500 French lentil grains per class
A10	Manual defect counts per lentil site per season	150 / 300 / 600	Not found; scales with trucks received	Ask a collector
A11	Arbitrations avoided per site per season	0 / 2 / 5	Not found; arbitration costs 130 € HT + 10 € [37] [V]	Ask a collector
A12	Wrong lentil allocations avoided per site per season	0 / 0.1 / 0.5	Not found; the key unknown behind A3	Pilot: compare decisions with and without the tool

9. Future extensions, ranked by evidence

"Strong" means market data and a paying precedent exist; "medium" means a documented problem but a thin premium or no precedent; "weak" means plausible only. None is certain until tested on French photos.

Extension	Evidence	Why	What it needs
Lentil reception grading: bruchid holes, broken, foreign material, pricing the sorting loss	Strong	~550 to 620 €/t at stake [NV]; highest damage of 2021-2025 in 2025 [15]; Cropify sells the same thing [26]	French lentil photos labelled by graders; calibration against visual counts [36]
Chickpea grading	Medium-strong	Cropify added chickpea in 2026 [26]; France grows 27,494 ha (2024) [13] [V] and is a net exporter	Calibre and defect classes; export specs
Food-soy intake check (green seeds, broken, stained, foreign seeds) for the six soyfood plants	Medium	Green seeds hurt the food outlet in 2025 [3]; concentrated buyers [6]; thin premium [10]	Buyer specs; a pilot with one plant
Faba bean for food export (Egypt <1 % bruchid)	Medium	Lost food export market 245 to 87 kt [21] [F]; strict thresholds [20] [V]	Photo undercounts hidden larvae: pair with X-ray or dissection sampling
Pea purity for fractionation (colour, broken, foreign seeds such as soy)	Weak-medium	Roquette grades colour, broken and purity [22] [F]; only ~30 €/t premium [25]	Purity classes, not holes
Producer-side check before delivery (our impact story: income and trust)	Medium	A grade the farmer can verify supports a fair réfaction [NV]; Track 3 asks about farmer income [1]	Same model; a farmer app flow; the farmer needs ~1,000 grains too
Packer QC before and after optical sorting	Weak-medium	Sorting is the standard fix [20] [V]; measures sorter efficiency [NV]	Access to a packing line
EU and other markets with the same crops	Medium	Cropify's own path: Australia, then Canada and the US [26] [V]	Local standards and data per market

10. What it takes to make it really useful

From what incumbents and regulators already require [35] [39] [28]:

- **French data:** photos from intake desks, several sites and at least two harvests, labelled by graders (agréeurs). R3 tells us how many grains per class.
- **Enough grains per sample** (table in section 5): about 1,000 grains, i.e. several photos, gives ±2 to ±3 points at lentil damage levels; plus a correction for the classifier's own errors.

- **Calibration against the reference method:** LABOSEM-style visual counts, daily check samples, monthly comparisons, as the USDA grain inspection service requires for Cgrain [35] [V].
- **Fixed capture:** a cheap tray or light box and a short list of phones. Inarix restricts its broken-wheat analysis to two phone models [28] [V].
- **Traceability and integration:** sample ID, stored images, results on the intake ticket; otherwise it stays a side app [NV].
- **Industry recognition:** Incograin certified an Inarix barley method [29] [V]; GTA judged Cropify worth "commercial consideration" [26] [V]. Arbitration still goes to a handpick reference [35] [V], so the tool starts as a pre-screen and evidence record.
- **Public funding for pilots:** the 2025 "Projets territoriaux filières légumineuses" call funded 10 projects for 11.7 M€ [42] [V]; France Relance put ~150 M€ into plant proteins [18] [V].
- **A channel:** partnership with an intake-desk incumbent, or the six soyfood plants as a first concentrated customer base.

11. Critiques we must answer

Critique	Evidence	Our answer
"Grading is not what holds French pulses back."	Sources cite price vs imports, yield risk, missing sorting tools and low demand (1.6 kg/adult/yr vs ~5 kg for the PNNS guideline) [16] [17] [18] [V]	True. We claim fairer pricing and trust at reception, not more demand. It supports farmer income (Track 3), not consumption.
"Labour savings are tiny."	Manual count ~2 to 4 € of labour [NV, 38]; lab impurity analysis 130 € HT per lot [37] [V]	Agreed. The value is decisions, disputes and speed (section 5), not labour.
"A better grade just moves money from farmer to collector."	If manual grading is unbiased, corrections cancel across trucks [NV]	Agreed. We do not sell € per point. Value comes from decisions that change and disputes avoided, and must be measured in a pilot [A3].
"One photo is too noisy for a 1 % threshold."	±6 points at 25 % damage with 200 grains; ~380 clean grains to show ≤1 % [NV]	Agreed: the product asks for several photos (~1,000 grains) and says "inconclusive" near the threshold.
"A photo misses bruchid larvae."	Pea: 3.8 to 4.1 % visual vs 11.8 to 13.5 % by dissection [36] [V]	We measure visible damage, the same thing the visual reference method counts, and say so.
"Food soy barely pays more."	<25 €/t gap [10] [11]	Correct. Soy is where public data exists to build and audit the method; lentil is the target, still untested on real lentils.
"Cropify already does it; Inarix could next season."	[26] [27] [29]	Proof that buyers pay. No technical moat: our edge is phone-only capture and being first on French soy and pulses, ideally with an incumbent as channel.
"2025 was an extreme year."	Lentil damage ~10 to 15 % in 2021-2024 [15]; soy green seeds 11 % in 2024 [3]	The 5 to 8 M€ figure is 2025 and an upper bound; a normal year is roughly half.
"Soy is animal feed."	~89 % of French soy is not food [2] [V]	We target the food-grade channel only; the guide excludes feed projects [1].

12. Still open (ask mentors or collectors)

- Current food-grade thresholds in Incograin addenda 7 bis (pulses for food) and 9 (soybean); members only [40].
- A price for downgraded or feed lentil, and any réfaction grid for bruchid-damaged or broken grains.
- How intake grading is really done for pulses: per truck or composite, by whom, how long.
- Official 2024-2025 lentil and chickpea tonnage.
- Whether Inarix plans pulse or soy defect analyses.

Sources

Opened on or before 31/10/2026. Full detail (what was read on each) in `research/viability/*.md` and `CONTEXT.md` §17.

[1] Hack the Fork, Participant Guide, Paris 2026 (event document).

[2] FranceAgriMer, Fiche filière Oléagineux 2026 (Feb. 2026).
https://www.franceagrimer.fr/sites/default/files/2026-02/18_FICHE_FILIERE_OLEAGINEUX_2026.pdf

[3] Terres Inovia, Fiche qualité des graines soja, récolte 2025 (Apr. 2026).
https://www.terresinovia.fr/sites/default/files/2026-04/fiche_qualite_graines-soja_2025.pdf

[4] Terres Univia, Chiffres clés 2024 (2025). https://www.terresunivia.fr/fichiers/publications/chiffres-cles-2024.pdf

[5] Eurostat, apro_cpsh1 (soya, France), queried 31/10/2026.
https://ec.europa.eu/eurostat/api/dissemination/statistics/1.0/data/apro_cpsh1?geo=FR&crops;=I1130

[6] Ortega (Terres Univia) and Lecomte (Terres Inovia), "Le marché français du soja", Spécial Soja 2024.
https://www.terresinovia.fr/sites/default/files/document_library/20126/437704/726505/special-soja-2024_vf.pdf

[7] Terres Inovia / ITAB, Guide de culture soja bio 2023, p. 21.
https://www.terresinovia.fr/sites/default/files/2025-06/guide-de-culture-soja-bio-2023.pdf

[8] Institut Veblen, FNH, Terres Univia, Mesures miroirs, cas d'étude soja (~2024). https://www.veblen-institute.org/IMG/pdf/fiche_soja_vf.pdf

[9] Réussir, Marché des oléagineux du 30/9 au 7/10/2026 (quotes Agreste 2026 forecast), 9/10/2026.
https://www.reussir.fr/articles/marche-des-oleagineux-et-coproduits-du-30-septembre-au-7-octobre-2026-les-prix-en-colza

[10] Terres Univia, Point éco 05-2025, Observatoire des prix payés aux producteurs de légumineuses.
https://www.terresunivia.fr/fichiers/publications/PE[05-2025]_web.pdf

[11] La France Agricole, "Les prix payés aux producteurs de légumineuses à graines ont progressé", 12/5/2025 (paywall, visible part).
https://www.lafranceagricole.fr/conjoncture-cultures/article/882292/les-prix-payes-aux-producteurs-de-legumineuses-a-graine-ont-progresse

[12] RAGT Seeds, interview of Y.-M. Milbeau (Olga), 17/1/2023. https://ragt-seeds.com/fr/soja-interview-de-m-yves-marie-milbeau-olga/

[13] Terres Univia, Statistiques Oléagineux & Plantes Riches en Protéines (2024 data).
https://www.terresunivia.fr/fichiers/publications/recueil-statistiques-2024-3.pdf

[14] FranceAgriMer, Fiche filière Protéagineux 2026 (Jan. 2026).
https://www.franceagrimer.fr/sites/default/files/2026-02/19_FICHE_FILIERE_PROTEAGINEUX_2026.pdf

[15] Terres Univia / Terres Inovia, Lentille, fiche qualité des graines, récolte 2025 (erratum).
https://www.terresunivia.fr/fichiers/publications/qualite-graines-lentille-2025-erratum.pdf

[16] Terres Univia, Consommation de légumineuses des Français: résultats Nutrimétrie (27/5/2026).
https://www.terresunivia.fr/l-interprofession/actualites/consommation-de-legumineuses-des-francais-resultats-de-letude-nutrimetrie

[17] Solagro / Réseau Action Climat, Les légumes secs: quelles initiatives territoriales? (c. 2015-2016).
https://agriculture.gouv.fr/sites/minagri/files/f77_publi-legumes-secs-web-2.pdf

[18] France Stratégie, évaluation France Relance, fiche Protéines végétales, Jan. 2024.
https://www.strategie-plan.gouv.fr/files/files/Publications/Rapport/fs-2024-rapport-france_relance_vol_ii_10_proteines_vegetales_0.pdf

[19] FranceAgriMer, communiqué 6/7/2026, appel à projets obtention variétale légumineuses. https://www.franceagrimer.fr/sites/default/files/2026-07/06072026_CP_FranceAgriMer_AAP_Obtention_vari%C3%A9tale_l%C3%A9gumineuses_2026.pdf

[20] Perspectives Agricoles n°467, June 2019, "Lutter contre les bruches des légumineuses" (Terres Inovia).
https://ecophytopic.fr/sites/default/files/2021-03/2019_06_PA467-68-bruches-legumineuses.pdf

[21] Réussir / La Dépêche, "Terres Inovia: le salut de la féverole passe par les OS", 8/9/2015.
https://www.reussir.fr/ladepeche/terres-inovia-le-salut-de-la-feverole-passe-par-les-os

[22] Réussir, "Les pois jaunes, une culture exigeante mais rentable", 4/1/2021.
https://www.reussir.fr/les-pois-jaunes-une-culture-exigeante-mais-rentable

[23] CRIT Grand Est, Référentiel de prix et de charges 2025 (DRAAF Grand Est), 16/4/2025.
https://draaf.grand-est.agriculture.gouv.fr/IMG/pdf/crit_16_avril_2025___annexe_2___referentiel_des_prix_crit_2025.pdf

[24] FranceAgriMer, prix payés aux producteurs trimestriels (lentil 873 €/t 2024/25, 806 €/t 2025/26). Source : FranceAgriMer.
https://visionet.franceagrimer.fr (SCR-CER-HISTOPRIXTRIMESTRIELS-C25-26.xlsx)

[25] Terres Inovia, Pois protéagineux, guide de culture 2025 (indicative prices 250 / 280 €/t). https://www.germineo.com/storage/cms/FT%20-%20Semences%20Ol%C3%A9agineux,%20prot%C3%A9agineux/POIS-PROTEAG.-2025-Guide-culture-TERRES-INOVIA.pdf

[26] ChannelLife, "Cropify AI lentil grading to go commercial this harvest", 20/8/2026.
https://channellife.com.au/story/cropify-ai-lentil-grading-to-go-commercial-this-harvest

[27] Stock Journal, "Cropify Opal AI tool grades grain in 5.3 minutes", 4/9/2026.
https://www.stockjournal.com.au/story/9341950/cropify-opal-ai-tool-grades-grain-in-53-minutes-boosts-farm-pay/

[28] Inarix, home, get-started and product pages, read 31/10/2026. https://inarix.com/

[29] Meta AI blog, "Inarix... DINOv2", 6/12/2024. https://ai.meta.com/blog/inarix-agricultural-supply-chain-meta-dino-v2/

[30] Ground Truth Ag, home page, read 31/10/2026. https://www.groundtruth.ag/

[31] The Western Producer, "High-tech grain grader wins people's choice award", 30/11/2023. https://www.producer.com/crops/high-tech-grain-grader-wins-peoples-choice-award/

[32] Cgrain, applications page, read 31/10/2026. https://www.cgrain.ai/applications

[33] FOSS, EyeFoss product page, read 31/10/2026. https://www.fossanalytics.com/en/products/eyefoss

[34] Next Instruments, SeedCount brochure (undated). https://cdn.labwrench.com/eMan/manualNo/10873/seedcount%20image%20analyzer%20brochure.pdf

[35] USDA AMS FGIS, Program Notice PN-26-03, Cgrain Value for whole kernels, 27/4/2026. https://www.ams.usda.gov/sites/default/files/media/PN2603_CgrainforWholeKernels.pdf

[36] F. Pavy (LABOSEM), "Analyse de grains bruchés", JIT Céréales et protéagineux, 10/3/2022. https://www.semae.fr/uploads/10-2022-jitcerpro_methode-grains-bruches_FP_CEK_MB.pdf

[37] Syndicat de Paris (Incograin), tarif des analyses (file 2021, linked in 2026). https://www.incograin.com/wp-content/uploads/2021/05/TARIF-SP.pdf

[38] France Travail job offers for seasonal silo agents (~2025), search snippets [F].

[39] Inarix help centre, "Data Enrichment: Master Sites", updated 24/4/2026. https://inarix.zendesk.com/hc/en-001/articles/32424124861213-Data-Enrichment-Master-Sites

[40] Incograin, Syndicat page (contract forms RUFRA, RULEGS; addenda 7, 7 bis, 9 members only). https://www.incograin.com/index.php/syndicat-incograin/

[41] Canadian Grain Commission, Official Grain Grading Guide, lentils, grade determination tables. https://www.canada.ca/en/grain-commission/services/grain-quality/official-grain-grading-guide/18-lentils/primary-export-grade-determination-tables.html

[42] Ministère de l'Agriculture, communiqué "11,7 M€... souveraineté protéique", 2/7/2025. https://agriculture.gouv.fr/telecharger/150164

[43] SoyMate dataset audits: CONTEXT.md §6, research/soybean_bruchid/findings.md, research/other_datasets/findings.md and new_candidates.md (Mendeley d97rjyk2tz, v6vzvfszj6, t94nzy4ptz, hwdtghttp7, c733bjz4m3; Radio_lentille on Hugging Face, JeanEudesH/Radio_lentille).

[44] Terres Inovia / Terres Univia, Pois, qualité des graines 2025 (Jan. 2026). https://www.terresinovia.fr/sites/default/files/2026-01/fiche_terres_inovia_qual_graines_pois2025.pdf

[45] Farm Online, "AI revolutionises pulse sampling with better accuracy on colour", 20/3/2025. https://www.farmonline.com.au/story/8921581/ai-revolutionises-pulse-sampling-with-better-accuracy-on-colour/

[46] Grain Central, Cropify endorsement article, 20/8/2026 (search snippet only, site blocked automated access) [F]. https://www.graincentral.com/?p=251800

[47] ZoomAgri: Grains West 2/9/2025 and press on pricing (snippets) [F]. https://grainswest.com/2025/09/ai-revolution-improves-analysis/

[48] Nebulaa MATT: Outlook Business and Business Today (~2018, snippets) [F]. https://www.outlookbusiness.com/big-idea/quality-check-4792